جامعة مصر للمعلوماتية
EGYPT UNIVERSITY
OF INFORMATICS

Egypt University of Informatics
Computer and Information Systems
Data Analysis Course

The Analysis of the Performance of Data Analysis Students

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Introduction

Academic performance is one of the most important indicators of student success, and understanding the factors that influence it is essential for improving educational outcomes. Among these factors, study habits—especially the number of hours spent studying—are often believed to play a major role.

This report investigates whether increasing study time leads to better academic performance (measured by GPA). Understanding this relationship can help students make better decisions about how to allocate their time and effort effectively.

Research Question

Hypothesis

- Null Hypothesis (H_0): There is a relationship between study hours and GPA.
- Alternative Hypothesis (H_1): There is no relationship between study hours and GPA.

Population of Interest:

- The population of interest is 102 of our university students.

Sampling Method:

- The data was collected using an online survey distributed among students. This represents a **convenient sampling method**, as participants were selected based on accessibility and willingness to respond.
- While this method is efficient and practical, it may not fully represent the entire student population. However, it is suitable for exploratory analysis and identifying general trends.

Bias Identification:

Several potential biases were considered:

- Self-reporting bias: Students may overestimate or underestimate their study hours.
- Selection bias: Only students who chose to respond are included.
- Response bias: Participants might answer in a socially desirable way.

To reduce bias:

- Questions were kept clear, neutral, and unbiased.
- No leading or suggestive wording was used.
- Responses were collected anonymously to encourage honesty and reality of data.

Survey Questions:

- How many hours do you study per week?
- What is your lecture attendance level?
- How regularly do you study?
- When do you usually start preparing for exams?
- What study method do you use?
- How many hours do you sleep per day?
- Do you have a job?
- What is your GPA?

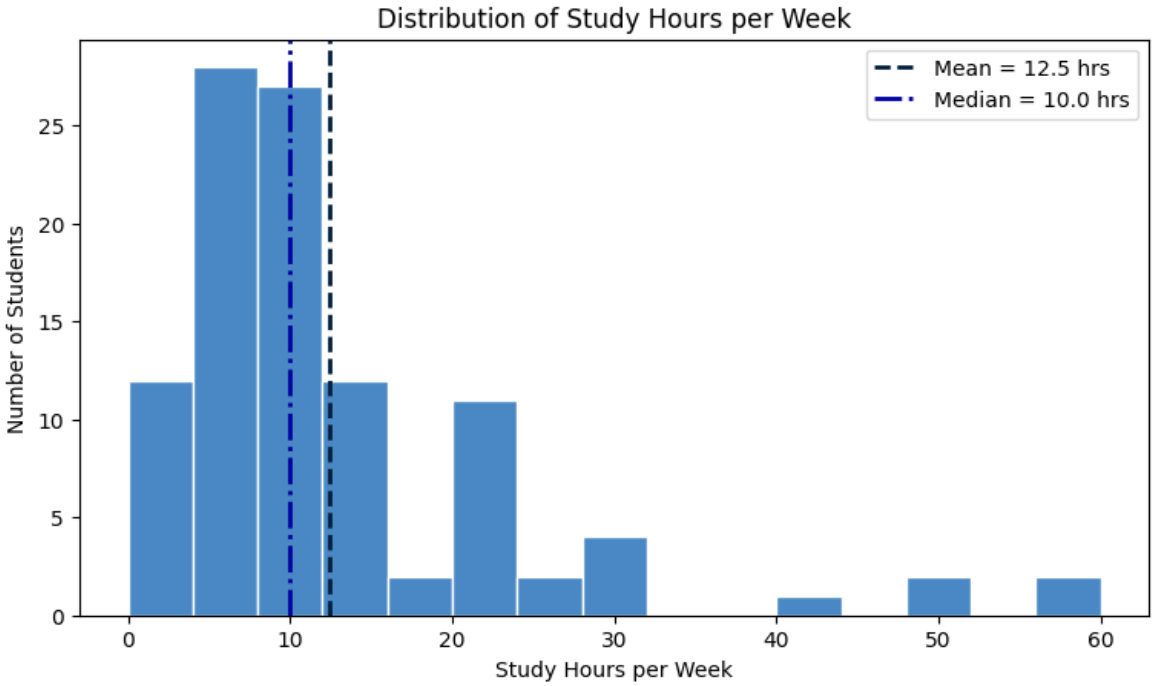
Online survey link: https://forms.gle/wW2AoHk5VWTJYrqeA

Number of samples collected: 102

Analysis:

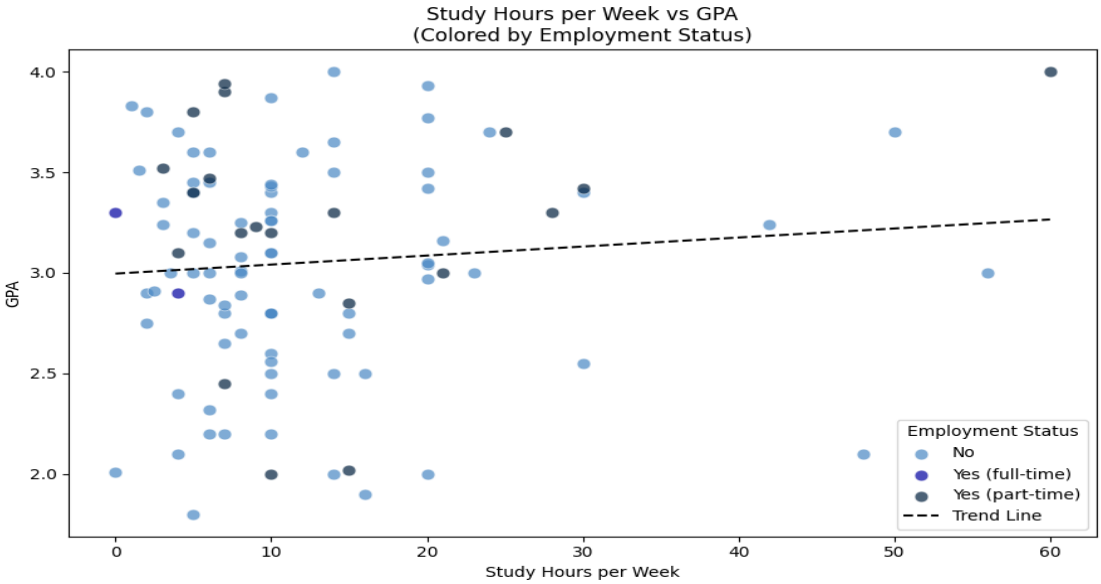
To analyze the data collected from this survey, we will calculate basic descriptive statistics such as the mean, median, and mode.

We will also create visual representations of the data using charts and graphs to help identify any trends or patterns.



Interpretation:

- The distribution is right tailed, meaning most students study between 5 and 20 hours per week, while a few study significantly more (up to 60 hours).
- The mean (≈12.5 hrs.) is higher than the median (10 hrs.) — this gap confirms the right skew caused by a small group of very high-study-hour students.
- The most common study range is roughly 5–15 hours per week.



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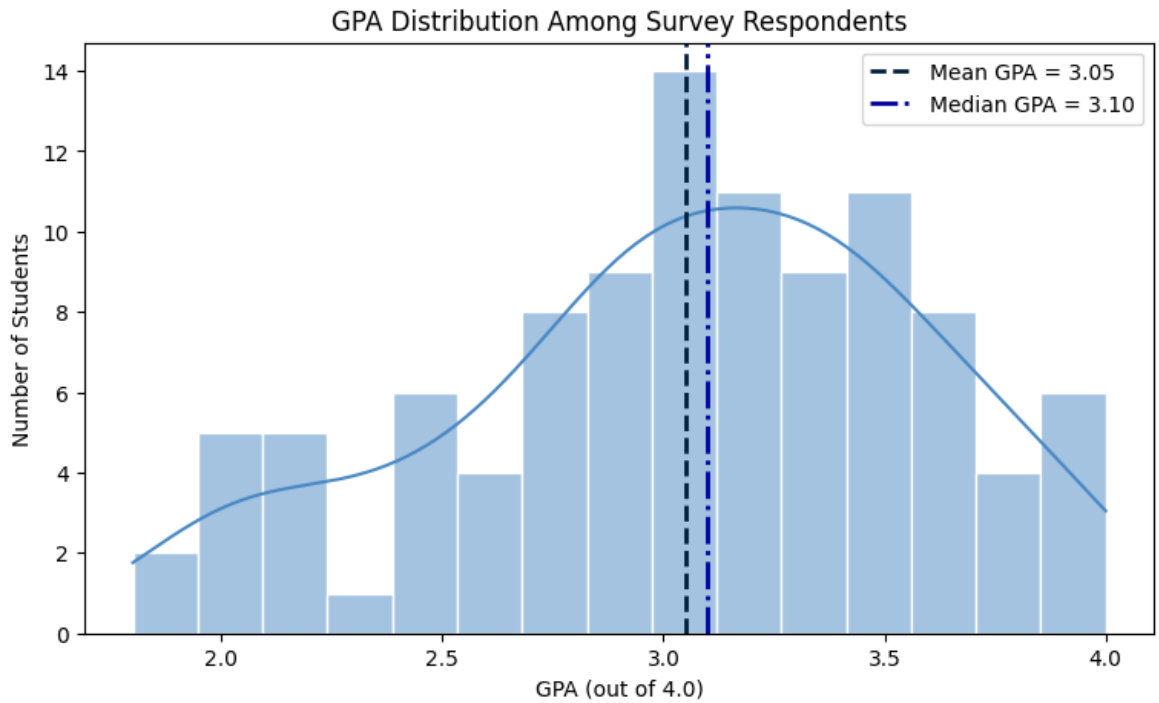
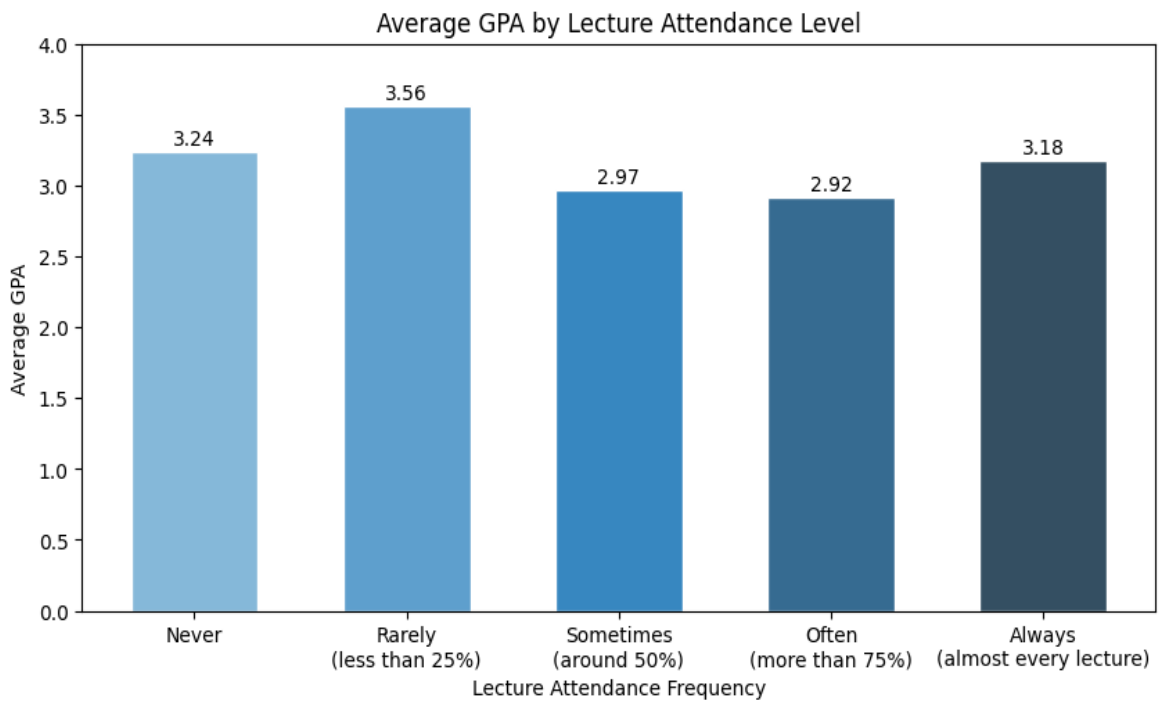


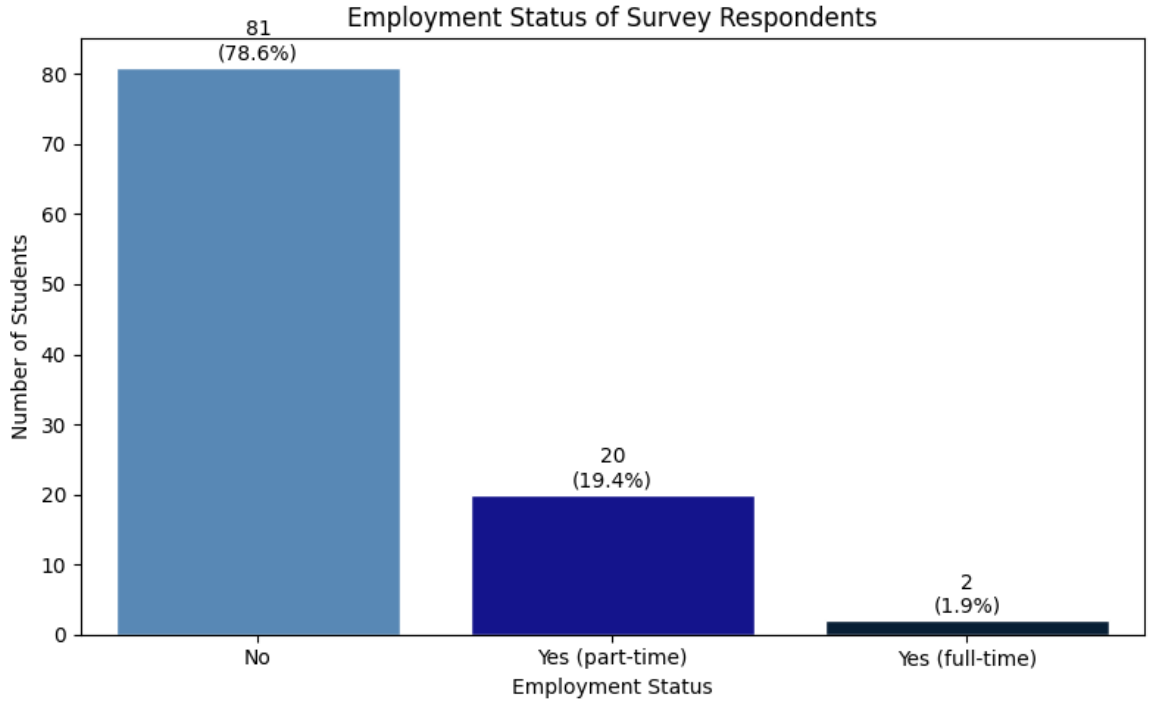
Interpretation:

- GPA follows an approximately normal distribution, centered slightly above 3.0.
- The mean and median GPAs are very close (≈3.08 and ≈3.10), indicating a balanced distribution without strong skew — most students perform in the B-range.
- Very few students fall below 2.0 or achieve a perfect 4.0, which aligns with typical university grade distributions.



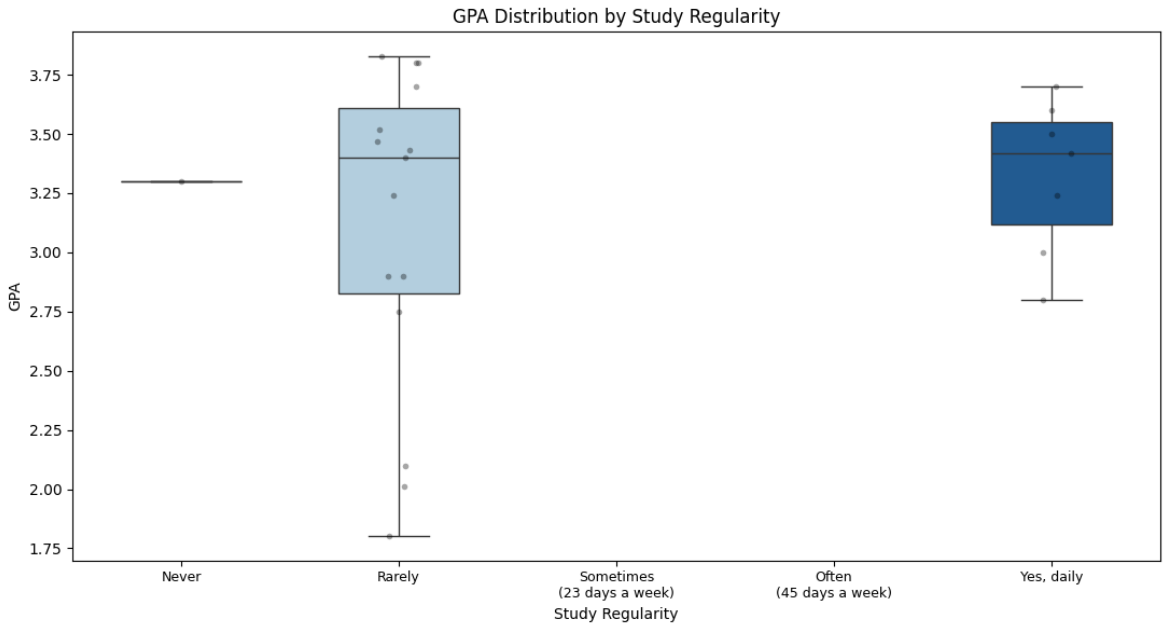
Interpretation:

- There is a general upward trend: students who attend lectures more frequently tend to have higher average GPAs, suggesting lecture attendance positively influences performance.
- Interestingly, "Rarely" and "Always" show competitive GPAs — indicating that some students who rarely attend lectures still perform well, possibly through self-study.
- Students who "Never" attend have the lowest GPA on average, confirming the value of in-person learning.



Interpretation:

- Most respondents (≈65%) are not employed, which is typical for full-time students.
- About 31% hold part-time jobs, balancing work and study simultaneously.
- Only a small number of students work full-time — these students may face the greatest challenge maintaining academic performance alongside demanding work schedules.



Interpretation:

- • Students who study daily ("Yes, daily") show a higher median GPA and narrower IQR, suggesting consistent high performance with less variability.
- Students who "Never" study have the widest spread and the lowest median GPA, indicating very unpredictable — and generally poor — academic outcomes.
- Moving from "Rarely" to "Often" shows a gradual GPA improvement, reinforcing the benefit of building regular study habits.

Conclusion

Based on the analysis, there appears to be a moderate relationship between study hours and GPA. However, the relationship is not strong enough to conclude that study time alone determines academic success.

Therefore, the results provide partial support for the hypothesis, but also indicate that academic performance is influenced by multiple factors beyond just study hours.

Critical Reflection (Potential Issues)

Several issues were identified during the data collection and analysis process:

- Outliers: An incorrect GPA value (e.g., 1111) was detected and corrected, indicating possible data entry errors.
- Sampling limitation: The use of convenience sampling limits generalizability.
- Self-reported data: Responses may not be fully accurate.
- Missing context: Factors like intelligence, course difficulty, and teaching quality were not included.

These limitations suggest that future studies should:

- Use larger and more diverse samples
- Include additional variables
- Apply more rigorous data validation techniques